

Human-LLM Pairing to Analyze Learning Outcomes for a Technology Ethics Course

Name	Name	Name
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Institution	Institution	Institution
City, State, USA	City, State, USA	City, State, USA
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Abstract

Course learning outcomes (LOs) define what learners should be able to demonstrate at the end of a course. However, the assessment of LOs can be time-consuming and challenging to do. To address this, being able to efficiently identify which LOs were salient and which were not is necessary. One way to assess LOs is through a bottom-up approach, analyzing course artifacts, such as student assignments, for evidence of learning outcomes being met. With the growing capabilities of large-language models (LLMs) to assist researchers working with large amounts of textual data, using an LLM to jump-start the process by generating an initial codebook can provide an efficient starting point. This poster presents an exploratory analysis of LOs by exploring course artifacts through human-LLM pairing for a course covering technology ethics broadly with undergraduate computing and engineering students. Using a hybrid approach, seventy-six (76) students' assignments were inductively analyzed. First, an LLM was used to generate a short list of themes from the assignments, which was used as the initial codebook, and then the researchers applied and extended the LOs by conducting a content analysis. Overall, the use of the LLM was valuable, though the themes required significant refinement by the researchers. We present the methodology and results from this analysis and highlight implications for this work.

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1 Introduction

Course learning outcomes (LOs) establish what learners should know and can achieve by the end of a course, but assessing overall course LOs effectively can be complex and time-intensive. With the growing capabilities of large-language models (LLMs), we aim to explore their potential for efficiently identifying course learning outcomes. This method could be especially beneficial for courses with multiple text-based assignments, streamlining outcome assessment. If this can be automated, it can provide both summative and formative feedback to students and instructors. LLMs are already being used for both teaching and educational support [1]. To this end, this preliminary work explores how human-LLM pairings can support the analysis of LOs from student assignments.

2 Methods

We analyzed data from seventy-six (76) students in an undergraduate technology ethics course. Students learned about technology, algorithms, and AI through case studies implemented as role-plays. In the final assignment, they were asked about each of the cases in the context of technology ethics. The four cases were about the Boeing Max crashes (Q1), facial recognition (Q2), AI in agriculture (Q3), and e-scooters on Campus (Q4) [2]. We analyzed 304 responses (76 participants x 4 questions). The average size of the responses was 76.8 words (SD = 46.4, median = 63, min = 21, max = 312).

2.1 LLM Codebook Generation

Following a similar methodology from [3], we prepared the data for codebook generation by 1) removing any identifiable information (names, locations, affiliations, etc.) and then breaking the data into sentence levels. Then, we used Ollama v0.1.0 with the Llama 2 model to run the codebook generation in a local environment. In the prompt provided to the model, the output requested was structured to include 1) a definition, and 2) keywords (Table 1).

Table 1: LLM Generated LO: Definition and Keywords

LO1	The rapid rate of technological change and its impact on societies around the world: globalization, surveillance, complexity, data, algorithms, artificial intelligence, AI.
LO2	The globalization of economies resulting from technology: corporate culture, shifts in global power, crime.
LO3	The changing nature of work: interconnected, governance, regulations, worker identity, reskilling, communications.
LO4	The creation of a global civil society: non-profit, not-for-profit, social benefit, social justice, sustainability, technology-for-good.
LO5	The development and transformation of the economy: future of work, personal identity, innovative technology.
LO6	The ethics associated with technology and professions: ethics, ethical decision-making, ethical dilemmas, professional ethics, code of ethics.
LO7	The social implications of technology: sustainability, loss of privacy, misinformation, fairness, accountability, transparency, bias, job-loss, black-box, equity, responsibility.

2.2 Analysis

Once the codebook was generated, the researchers conducted a content analysis exploring both the codebook that was generated as a starting point and the student responses to the assignment.

They first read through the initial codebook, exploring how the LLM had split and navigated the outcomes. Then, they read through a random selection of 150 responses. Next, they tried to apply the outcomes that were generated and noted any exceptions or questions. They met and discussed revising, combining, and modifying the outcomes to better analyze the dataset. Then, they applied the new learning outcomes (Table 2) to the dataset.

Table 2: Final LO: Definition and Keywords

LO1	The rapid development of technology and its impact on societies around the world: globalization, innovation, complexity, transformation, shifts in global power, crime.
LO2	The changing nature of work: automation, job loss, future of work, worker identity, human skills, reskilling, standards, governance, regulation, corporate culture, laws.
LO3	The creation of a beneficial civil society: not-for-profit, social benefit, social justice, sustainability, technology-for-good, safer-society, beneficial impact, respect.
LO4	Discussion about role-play activities and case studies: developing software, surveillance, chatbot, sensors, drones, facial recognition, biometric, green energy.
LO5	Technology and ethics: ethical decision-making, ethical dilemmas, professional ethics, code of ethics, philosophy, virtues, moral, utilitarian, ethical gray area.
LO6	The effects of technology: privacy, trust, misinformation, rights, fairness, black-box, bias, accountability, transparency, equality, equity, responsibility, inclusive, consent, environment, marginalized.

For inter-rater reliability, *Krippendorff's Alpha* (multiple raters that can rate an item 0 or more categories) was calculated at 0.689.

3 Results and Discussion

We present the results from the LLM categorization (Figure 1) and the human categorization (Figure 2). Notably, the LLM categorization tended to fixate on specific words. Because of the assignment question, students would reference "ethics" often, and the LLM was influenced by this. Additionally, even when provided the option to give no category, the LLM would try to identify a category. However, in many cases the detail was just surface level.

Regarding the revised analysis, the researchers noted that the LLM codebook provided a great starting point to visualize what kind of data was present. The human analysis was more nuanced both at a micro level regarding words like "ethics" and also in providing a broad set of themes. The researchers also noted there was some ambiguity in the initial LLM codebook keywords, but they were refined down, and the keywords were made more separate over the initial discussions among researchers.

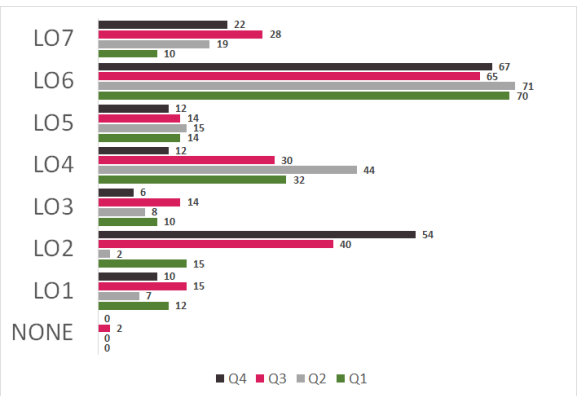


Figure 1: LLM Approach

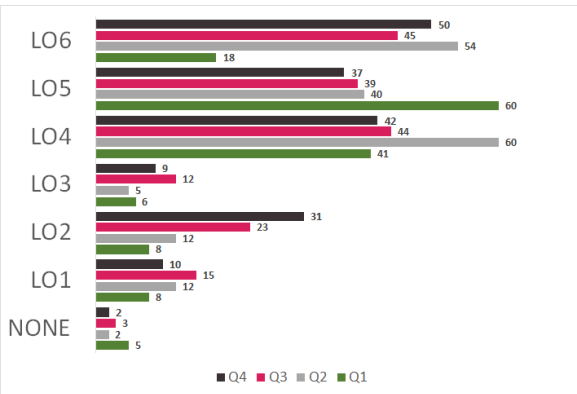


Figure 2: Human Approach

4 Conclusion and Future Work

In this preliminary study, we found that the use of the LLM to generate the initial codebook provided a sufficient starting ground. Future work in this realm could explore providing this type of analysis in a timely and easily accessible manner for instructors. This can yield benefits in two ways: feedback can be provided to students that directly connects the assignments to the desired outcomes of the course. Additionally, instructors can receive feedback on if any of the outcomes are not being met and adjust their teaching accordingly.

Acknowledgments

Anonymous for review.

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